Pierros Ntelis | Curriculum Vitae

Personal Information	Born on September 18th, 1990 in Athens, Greece Greek citizenship
Contact	NEWMILU, e-mail: ntelis.pierros -at- gmail.com
Research interests	Mathematics, cosmology, theoretical physics, high energy astrophysics, large scale structures - theoretical and observational aspects
Academic	2026-Now School of Physics, Harbin Institute of Technology, Visiting Professor 2025-Now Theoretical Physics at National University of Uzbekistan, Tashkent Senior Researcher (equivalent to Associate Professor) 2022-2026 **International Bilingual School of Provence** Mathematics and Physics Teacher for IB, IGCSE 2018-2020 **Center of Particle Physics of Marseille, Aix University** C.N.E.S. Postdoctoral Research Associate 2014-2017 **Astroparticle and Cosmology Group, Université de Paris** PhD thesis: *Probing Cosmology with the homogeneity scale of the universe through large scale structure surveys*, supervised by Dr. J.C. Hamilton 2013 **National Technical University of Athens** Diploma-Thesis: *Hawking Radiation* supervised by Prof.K.Anagnostopoulos, (6 months) 2012 **National Observatory of Athens** Internship: *Calculation of the age and chemical compositions* of star clusters, supervised by Dr. A. Dapergolas, (2 months)
Education	2014-2017 **Ph.D.**, **Université de Paris, Denis-Diderot** Fundamental Physics, Cosmology, 1st upper class honours 2013-2014 **M.Sc.**, **University Paris 6, Paris 7, Paris 11** Nuclei, Particles, Astroparticles & Cosmology, Grade: $14.9/20$ 2008-2013 **B.A.**, **National Technical University of Athens** School of Applied Mathematical & Physical Science, Grade: $8.5/10$ 2002-2008 **Lyceum of kareas**, **Athens**, Grade: $18.7/20$
Languages	English (fluent), French (intermediate), Greek (native)
Memberships	05/2024- , CosmoVerse network, Cosmic WISPers network 06/2025- , Fundamental challenges in physics theories network 2018-2021, Euclid Consortium, DESI Survey 2014-2017, SDSS-III, BOSS Survey
Awards and Fellowships	2024-2025, 2 CosmoVerse STSM grants E-COST-GRANT-CA21136, (2 week) 2024-2025, Cosmic Whispers YRIG grant E-COST-GRANT-CA21106, (1 week) 2018-2020, C.N.E.S. Post Doctoral Fellowship, (2+1 years) 2014-2017, STEP-UP UnivEarth Sciences, PhD Fellowship 2014 , Paris Denis-Diderot University, Internship Fellowship, (2 months) 2013-2014, Master Award Fellowship, J.S. LATSIS FOUNDATION, (1 year) 2012 , IKY E.S.P.A., Internship Award Fellowship, (2 months)
Publications	ArXiv, ADS, I-HEP, RG, Linkedin, COSMOPIT, ORCID, GS, SP, zM, RL, UT, Z
Interdisciplinary Skills	Maths, efficient communicational skills and learning, data and stat analysis, Mathematical Physics, AI, Python, C/C++, Fortran, Mathematica, LaTeX, HPC, Unix/Linux, CAMB, CLASS, COSMOPIT, mangle, Euclid softs

Teaching & Academic Services	2026, HIT, Harbin, A note on data analysis using Bayes' inference (code), (1h)
	2025, CPT at AMU, Marseille, Advanced Math methods in cosmology, (1h)
	2025, Auckland Univ., New Zealand, Advanced Math methods in cosmology, (1h)
	2025, CPPM, France, Advanced Math methods in cosmology, (1h)
	2025, New U.U., Uzbekinstan, Advanced Math methods in cosmology, (1h)
	2025, SSM, Napoli, STSM: Tensions and Non-Local Analytical Dynamics, (1w)
	2024, ISSA, Malta Univ., STSM: (Non-)Riemannian Cosmic Tensions, (1w)
	2024, Perimeter Institute, Celestial Holography Summerschool participant, (1w)
	2024, Cosmo From Home, Session host in Dynamical System in Cosmology, (1h)
	2023, UCO, Benasque, Host in Modifications of Gravity and Higher-order stats, (5h)
	2023, Cosmo From Home, Session host in Homogeneity, Simulations and LSS, (5h)
	2022, IBS, Mathematics and Physics for High School Students, (6$\times$4 h Seminars, 3y)
	2021 , Euclid , Science talk: Overview of impact of interlopers on PNG, (1h Seminar)
	2020 , Aix-University, From Horndeski to AofEFT, (40' Seminar)
	2020 , Aix-University, M2 intern supervisor on LSS Diagnostics, (25 hours)
	2020 , Aix-University, M1 intern supervisor on BAO/neutrinos, (3 month)
	2020 , Aix-University, M1 intern repporter on Strong Lensing, (1h exam)
	2020 , Aix-University, M1 intern supervisor on BAO, (1 month)
	2020 , DESI, Participation on spectra Visual Inspection (2 months), (2nd Prize in SR)
	2019 , Aix-University, Homogeneity with Multitracers, (2h Seminar)
	2019 , Aix-University, A note on lss, Summer School, (6h Seminar)
	2018 , Renoir, co-organiser of science meetings "Show Plot Club", (weekly, 3y)
	2018 , LSS survey internal reviewer service (DESI, Euclid) (3y)
	2018 , CPPM, LAM, Participation on QC tests of NISP Euclid instrument, (6 months)
	2018 , Aix-University, Scientific Watch, (6h Seminars)
	2018 , SEMFE, NTUA, *Observational Cosmology*, (1.5h Seminar)
	2018 , UiO, What can fractality tell us about the universe?, (1h seminar)
	2018 , CPPM, M1 intership supervisor on BAO, (2 months)
	2018 , Aix-University, A note on lss, Summer School, (3h Seminar)
	2017 , SEMFE, NTUA, *Exploring cosmic homogeneity*, (1.5h Seminar)
	2016-, Referee Report, MNRAS journal
	2015 , Paris-Denis Diderot, Statistical Methods, (4 months)
	2014, APC, Paris 7, co-organiser of cosmology journal club, (weekly, 3y)

Conferences Talks/Posters	2025, The Dark Side of the Universe, FAT dynamics to cosmology (25' Talk)
	2025, Cosmo From Home, Advanced math in cosmology (25' Talk)
	2024, 50 yr of Horndeski at PI, A probabilistic expanding universe under FA, (Poster)
	2024, Cosmo From Home, A probabilistic expanding universe under FA, (25' talk)
	2023, NEB, New avenues of functors of actions theories, (20' Talk)
	2023, Cosmo From Home, New avenues on Functors of Actions, (25' talk)
	2023, NTUA HET, New theories in cosmology and particle physics, (1h Seminar)
	2022, CERN TH DPT Visit, Functors of Actions (5' Blackboard Talk)
	2021, Euclid Science Talk, Overview of impact of interlopers on PNG (1h Talk)
	2020, Atelier D.E., PNG with interlopers in Euclid, (20' Talk)
	2020, Atelier D.E., Perturbation Theory of LSS, code: CLASS-PT, (20' Talk)
	2020, SEMFE, NTUA, LSS: Theory and Observations, (40' Seminar)
	2019, 30th SYMPOSIUM, Fractality from Multiple Tracers, (15' Talk)
	2019, Euclid IAS Paris, Systematics on PNG, (20' Talk)
	2019, Euclid Consortium Helnsinki, Primordial Non-Gaussianity, (15' Talk)
	2019, eBOSS/DESI France, Cosmology with Cosmic Homogeneity (15' Talk)
	2018, Euclid GC-WL, Ameliorating cosmology with homogeneity, (15' Talk)
	2018, Colloquium of D.E., Cosmic Homogeneity as a standard ruler, (20' Talk)
	2018, R. Moriond, *Euclid:Homogeneity in search of the Dark Sector*, (Poster)
	2017, P.O.N.T. in Cosmology, *Is the universe homogeneous?*, (20' Talk)
	2016, R. Moriond, *The homogeneity scale of the universe*, (15' Talk)
	2015, UnivEarthS School, Santorini, *LSS and homogeneity*, (Poster)
	2015, Advance School of Cosmology at I.C.T.P., Trieste
	2015, Rencontres de Jeunes Physicienne, *Homogeneity* (Video), (15' Talk)
	2015, IDPASC school, at LPNHE, Paris
	2014, Elbereth Conference, IAP, *The homogeneity of the universe* (15' Talk)
	2014, UnivEarthS School, *The homogeneity of the universe*, (3' Talk)
Certificates	Qualif. of « Maitre de conférences » (sec. 26 "Applied Math") obtained 2026
	Qualif. of « Maitre de conférences » (sec. 29 "Elementaries") obtained 2019
	IB Mathematics: Applications and Interpretations
Outreach Talks	2024-Today, Utube, Introduction cosmology for everyone (1h talk)
	2023('24), IBS of Provence, Neuroscience and cosmology for high school students (3h)
	2023 , IBS of Provence, Math and physics from CPT for high school students (3h)
	2022 , IBS of Provence, Mathematics and physics for high school students (3h)
Other Interests	creative writing, piano, guitar, running, water sports, skiing

Refs. Dr. J.C. Hamilton
APC (PhD Supervisor)
5 Rue Thomas Mann
75013, Paris, France
tel.: 00 33 1 57 27 69 26
hamilton@apc.in2p3.fr

Prof. Jackson Levi Said
UM-ISSA
Mathematics and Physics
University of Malta, Msida
tel.: +35 62 34 03 035
jackson.said@um.edu.mt

Prof. S. Camera
INFN
via Pietro Giurra, 1
Torino, Italy
tel.: 00 39 1 43 87 44 07
stefano.camera@unito.it

Prof. S. Capozziello
SSM Institute
Largo S. Marcellino, 10,
80138 Napoli, Naples, Italy
tel.: 00 39 329 909 4346
salvatore.capozziello@na.infn.it

Prof. S. Katsanevas
EGO (Mentor)
Via Amaldi 56021 Santo Stefano
a Macerata Cascina (Pisa), Italy
tel.: 00 39 0 50 75 23 00
stavros.katsanevas@ego-gw.it

Dr. A. Ealet
IPNL
4 Rue Enrico Fermi
69622 Lyon, France
tel.: 00 33 6 42 25 44 00
a.ealet@ipnl.in2p3.fr

Prof. Bobomurat J. Ahmedov
NUU (Mentor)
Institute of Theoretical Physics
100174, Tashkent, Uzbekinstan
tel.: 00 998 71 2358102
ahmedov@astrin.uz